Detecting Long Period Transients with VLITE

VSPARC: The VLITE Survey of Polarized Radio-transient Candidates

Aarushi Kanigicherla
Sarah Trainer

Thomas Jefferson High School for Science and Technology
Astronomy and Astrophysics Laboratory

Lab Director: William D. Linch III

Principal Investigator: Emil Polisensky

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Abstract

Objects called long period transients (LPTs) are a recently discovered phenomenon of high peaks in radio emission before a subsequent nulling period [13]. The origin of these phenomena is unknown, largely because only 24 have been discovered thus far. Our goal is to expand this dataset in an effort to develop a solid understanding of these bodies.

The Very Large Array (VLA) Low-band Ionosphere and Transient Experiment (VLITE) provides low frequency (320–384 MHz) observations at a nearly continuous rate [10]. Three LPTs have already been corroborated with VLITE data, and there is still more data to analyze. We aim to use VLITE data to find new LPTs in an effort to glean new information about these mysterious radio phenomena.

We will create algorithms to filter out known radio sources and remove image artifacts. We will then check our candidates by hand in an attempt to discover new examples of these mysterious astronomical bodies.

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Contents

1 Introduction

Our galaxy is filled with various radio emitters, from pulsar binaries to radio galaxies. This work describes attempts to detect a new class of astronomical objects known as long period transients (LPTs). These mysterious sources emit bright, highly polarized radio pulses lasting minutes to hours, followed by long intervals of silence that can span days, months, or even years [11]. Their rarity and unusual properties make them scientifically valuable, but also difficult to detect with conventional observing strategies. Recently, researchers reported ASKAP J144834−685644, a new LPT with a $\sim$1.5 hour period that was observed in radio, optical, and X-ray. This multiwavelength detection is rare among LPTs and suggests that some of these objects could be linked to magnetic white dwarf binaries or other stellar systems [1]. Findings such as these highlight how little is known about LPTs and their properties. With only a dozen or so well-characterized LPTs, it is difficult to establish population-level trends in pulse duration, repetition rate, polarization fraction, spectral behavior, or multi-wavelength counterparts [17].

From an observational perspective, LPTs occupy a previously scarce region of the radio transient plane, which maps pulse duration against luminosity [11]. They lie between fast pulsars and millisecond transients and slower, incoherent phenomena on the other. This positioning helps explain why LPTs went undetected for so long: wide-field surveys often lack the cadence or sensitivity to capture rare, minute-to-hour long while targeted pulsar searches typically assume much shorter periods. Most known coherent radio emitters, such as pulsars, rotate rapidly with periods ranging from milliseconds to seconds. In contrast, LPTs have periods ranging from several minutes to hours, far slower than what standard neutron-star emission models predict to be possible [11].

This paper focuses on analyzing data collected by the Naval Research Laboratory's Jansky Very Large Array (VLA) Low-band Ionospheric and Transient Experiment (VLITE). VLITE's receivers are currently installed on 18 of the 27 antennae on the VLA in New Mexico, detecting radio frequencies with a bandwidth of $\sim$40 MHz centered at $\sim$340 MHz while the VLA simultaneously collects data at higher frequencies [10]. As a result, VLITE collects more than 6000 hours of imaging data each year, much of which remains in need of analysis. Because LPTs emit radio waves, VLITE is particularly well-suited for their detection. Its typical sensitivity of $\sim$0.05–0.2 Jy for 10-minute pulses is well matched to the observed flux densities of known LPTs. Importantly, VLITE has already detected 3 of the $\sim$13 well-studied LPTs reported in the literature, corresponding to a recovery rate of roughly 30%. The detection of multiple known LPTs in archival data suggests that further investigation has potential to yield additional candidates. Identifying an existing LPT would also be of merit, as it solidifies the effectiveness of our methodology. Ultimately, increasing the LPT sample size will allow for stronger classification and analysis of properties.

Our research utilizes VLITE source catalogs from imaging data across the A, B, and C configurations, examining VLITE images to identify candidate LPTs. We first filter the data to reduce the number of artifacts and false detections based on properties such as high signal-to-noise ratios. Sources are also filtered for compact morphology at arcsecond scales to remove extended sources such as supernova remnants. Candidate sources are then cross-matched against existing radio and multiwavelength catalogs to exclude known sources and visually examined to eliminate image artifacts; any remaining sources will be examined for properties characteristic of LPTs, such as large-amplitude variability on minute-to-hour timescales, strong linear and/or circular polarization, and irregular detection consistent with long nulling intervals. Throughout the process, we compare our filtered sources to known LPT detections to validate our selection methodology and prevent the filtering process from removing probable true detections.

The broader scientific impact of this work extends beyond LPTs themselves. LPTs share traits with other classes of radio transients, including pulsars, rotating radio transients, magnetically active stars, and some subclasses of fast radio bursts, or FRBs [11]. Improving techniques for imaging, filtering, and classifying VLITE transients will strengthen the methodology used to study radio variability across the sky. Additionally, VLITE's cadence and commensal observing mode resemble those planned for next-generation facilities such as the Square Kilometre Array (SKA), making this project valuable for future transient-search pipelines. Additionally, a systematic search for LPTs within VLITE data may lead to serendipitous discoveries beyond the primary focus of the study. Cross-matching VLITE detections could identify stars previously thought not to emit in radio wavelengths, as well as potentially discovering unexpected transient or variable sources.

1.1 The Nature of LPTs

LPTs are elusive objects. Although the first LPT-like source (GCRT J1745–3009) was identified in the early 2000s, only within the past several years has a small, but diverse, population been uncovered and studied in detail [13]; as of this writing, only 24 have been discovered thus far. Their true nature evades specific classification. LPTs will "turn on" in bursts that range from minutes to hours, a range that has historically been poorly investigated because of its unoptimal range for study from radio data [11]. Their discovery is complicated by the fact that LPTs will often turn off in long nulling periods that can last years. Additionally, most known LPTs are close to the galactic center or in otherwise crowded areas, making it difficult to observe them in wavelengths other than radio and gather further information about their properties. The search for LPTs is, however, made somewhat easier by their high levels of linear and circular polarization. Current observations indicate that LPTs are highly polarized, sometimes approaching 100% linear polarization, in which the electromagnetic radiation is oscillating almost entirely in the same plane, and often show polarization mode switching or linear-to-circular conversion [6, 11]. LPT emission also tends to be coherent (the phase difference between individual waves remains constant over time), which enables us to further filter the large amounts of data and more effectively verify candidates.

Theories regarding the nature of LPTs vary — it is even uncertain whether they are isolated objects or binaries — but the most common theories involve white dwarf/M dwarf systems and neutron stars [1]. Their high polarization and coherent radiation strongly implicates emission from organized, highly magnetized environments, such as neutron-star magnetospheres or white dwarfs with strong magnetic fields. Some LPTs have been found with companions visible in non-radio wavelengths; such LPTs may be part of a binary system with a white dwarf, though not all LPTs have detectable companions. In sources that show compelling evidence for white dwarf and low-mass star binaries, radio emission might be powered by magnetic interaction or orbital beat frequencies rather than rotation alone. Namely, some LPTs may be part of a special class of magnetic cataclysmic variables known as polars [1], a binary system in which an extremely strong magnetic field around a white dwarf pulls in material from a companion overflowing its Roche lobe. The bursts, then, would originate from the shock of this material accreting onto the white dwarf's magnetic poles. However, other sources show properties more consistent with neutron stars or magnetars, including pulsar-like polarization swings, glitch-like events, or phase-locked radio and X-ray emission [11]. Some LPTs appear similar to neutron stars, but in most cases their periods are far too long for the pulses to be from pulsars. According to the traditional "pulsar death line," the point in the neutron stars spinning as slowly as the characteristics of LPTs would imply should not have the voltage required to generate coherent radio emission [4, 17]. Yet LPTs violate this, emitting radio pulses with brightness temperatures exceeding 10^{16} K, which must be explained by a coherent emission process (such as the electron cyclotron maser, or ECM) rather than by simple thermal radiation [13].

Lack of consensus about the nature of LPTs highlights a gap in our understanding of these systems, but also reflects variability in the data: identified LPTs differ greatly in their characteristics and behavior. Even one of the most distinguishing characteristics of LPTs, their long periods, can change from source to source. Some objects have periods as short as 7 minutes, such as CHIME J0630+25, while others extend to over 6 hours, such as ASKAP J18390756. This variability suggests the possibility that multiple physical mechanisms are at play or that existing models of magnetosphere emission are incomplete. As it stands, the classification for LPTs is relatively broad and based almost entirely on the fact that they 1) emit radio waves and 2) have periods longer than a few minutes [17].

2 Methods

Analysis took place across data from VLITE's A, B, and C configurations, which have baseline lengths of roughly 36, 11, and 3.5 km and resulting resolutions of 5, 15, and 45 arcseconds respectively [10]. The D configuration was omitted from our search due to its low resolution—detecting point sources can be difficult in the D configuration, as it is harder to distinguish between dim point sources and imaging artifacts. For each configuration, we began with two processed catalogs of sources, one with sources emitting in the Stokes I band (I) and the other containing polarized emissions (P). In addition, we had I and P image data catalogs containing information on noise levels, beam parameters, observing time, etc. The source catalogs were assembled through the Python Blob Detector and Source Finder (PyBDSF), which detects isolated sources within FITS files and allows for these sources to be further investigated without manually searching every image [15]. These catalogs contain sources in the galactic plane, where most other LPTs have been found. The I catalogs were made using standard PyBDSF thresholds ($> 3\sigma$ island boundaries and $> 5\sigma$ peak detections), while the P catalogs required higher thresholds (5-7σ), due to the non-Gaussian noise characteristics of images taken in polarized light. This approach reduces false positives, but also may limit sensitivity to low-level polarized emission.

Each configuration contained detections of at least one known LPT: the A configuration contained one detection of one LPT, the B configuration dataset contained seven detections of two LPTs, and the C configuration data contained five detections of two LPTs. These detections were used as benchmarks throughout the filtering process to determine the effectiveness of the cuts we made throughout.

2.1 Filtering LPT Candidates

Initial catalog analysis was performed by matching the I catalog with data from the GAIA mission to determine if any new radio stars were detected within 50 parsecs of the Sun; since GAIA images in optical and our data is in radio wavelengths, matches would indicate novel detections of radio emissions from nearby stars [7]. We constructed our dataset from GAIA mission Data Release 3 (DR3) by querying ESA's GAIA archive with the astroquery.gaia interface[8, 9]. We configured the parallax parameter as greater than 20 the parallax_over_error parameter as greater than 10 to ensure that any inferred erroneous parallax distances from GAIA scatter were minimized in our GAIA dataset. Then, we used TOPCAT's crossmatching feature with the "All matches" parameter to match celestial coordinates of objects in the GAIA catalog and in our Stokes I catalog with a separation of 20 arcseconds. This resulted in 40 spatial matches across our Stokes I catalog and GAIA data.

In order to ensure a higher likelihood of real detections post-filtering, we selected sources within the I catalog for a signal to noise ratio (S/N) higher than 10 (where bm is the solid angle covered by the antenna beam in each image) and a compactness more than 0.9. The S/N cut ensures that the detected source is at least ten times brighter than the surrounding noise, which is an indicator of the detection's veracity—between an S/N of 5 and 10 detections are highly ambiguous, and below an S/N of 5 detections are likely not real. The compactness filter selects our data for bright point sources (like LPTs) rather than more extended radio emitters such as supernova remnants or radio galaxy lobes. To further remove extended sources, we also removed any detections cataloged as "M", indicating that they represent multi-component sources which are not pertinent to our study.

We next performed exhaustive filtering to remove as many known sources as possible from the I catalog, as true LPT detections would likely not be represented in known radio catalogs. Using the Astropy coordinate matching features, we calculated the separation between each source in the catalog and known sources in several radio catalogs: the NRAO VLA Sky Survey (NVSS), the TIFR GMRT Sky Survey (TGSS), The H I, OH, Recombination line survey of the Milky Way (THOR), the Rapid ASKAP Continuum Survey (RACS), catalogs of GHz-Peaked Spectrum radio sources (often young radio galaxies) near 1 and 5 GHz (GPSR1 and GPSR5), and catalogs of known pulsars [2]. As sources with a separation to a known source below the resolution of the configuration (15" for B, 45" for C) were certainly known sources, those were discarded. Additional cuts beyond the resolution were performed to reduce the size of the dataset. Full histograms of the separations and the resulting separation threshold cuts used for each catalog can be found in Appendix B.

In the final phases of our filtering, we removed sources that were detected in the same image as other sources that passed the catalog separation, as those have a higher likelihood of being artifacts of a bright source; such artifacts would appear around the source and typically result in more than one detected source in the same image. The remaining detections have a higher likelihood of being sources in their own right, and not artifacts.

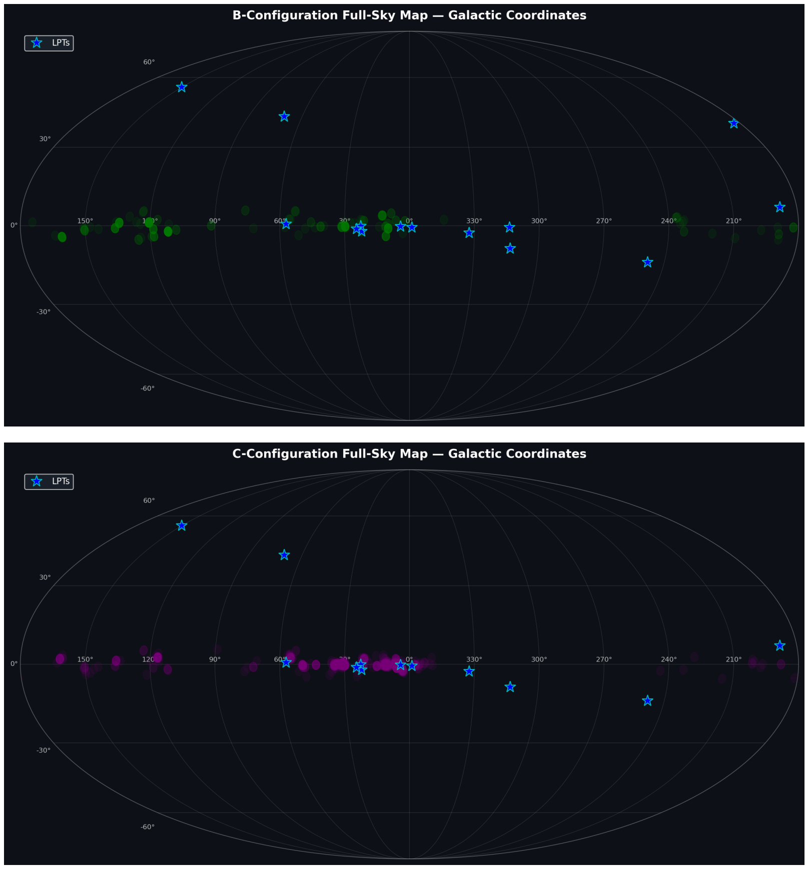

Figure 1: Final B configuration (top) and C configuration (bottom) candidate LPTs overlaid with detections of known LPTs.

2.2 Image Analysis

Our images were processed using the PyBDSF algorithm to detect sources before being shared for visual analysis with DS9 [15]. In order to simplify the inspection process, which often took place across hundreds of images, the original FITS files were modified to place the coordinates of the detection in question at the center and crop the image to a smaller region around it. The main goal of visual inspection was to differentiate between true sources and sidelobes, spots of brightness produced as an artifact of the imaging technique. In VLITE data, due to the Y-shape of the

VLA, such sidelobes fall along a six-pointed star-like symbol; this shape is formed from each axis of the Y-shaped array, intersecting at the source and continuing past it. This phenomenon is seen quite obviously in the false source depicted in Figure 2, a sidelobe of the bright source to its northwest. This sidelobe is quite close to the source that generated it; many sidelobes are farther away from true sources, and as such harder to differentiate from true detections without careful inspection.

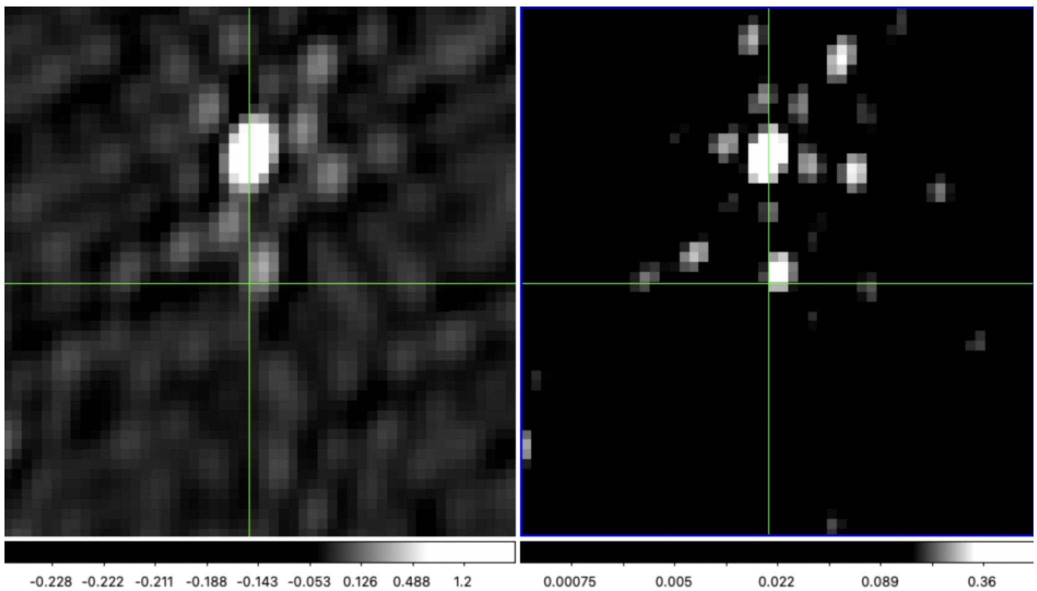

Figure 2: Intensity (left) and polarized (right) images of an identified sidelobe. The crosshair is located at the bottom left corner of the false source.

Such sidelobes contrast with true detections, which are most identifiable by the sidelobes they themselves generate; see Figure 3 for an example of a true LPT detection in the C configuration.

In order to quickly identify sidelobes, especially when they are located farther away from their parent source, the visibility functions of known sources, calculated from the object's brightness, size, distance, etc., were subtracted from the image. This removed interference patterns and artifacts thrown by those sources such that sidelobe detections were removed from the images altogether. Images with visible sources after this sidelobe removal process were examined in their corresponding P catalog image to identify candidate LPTs; since LPTs typically emit both polarized and nonpolarized light, a credible detection in both the I and P catalogs would signal a robust candidate.

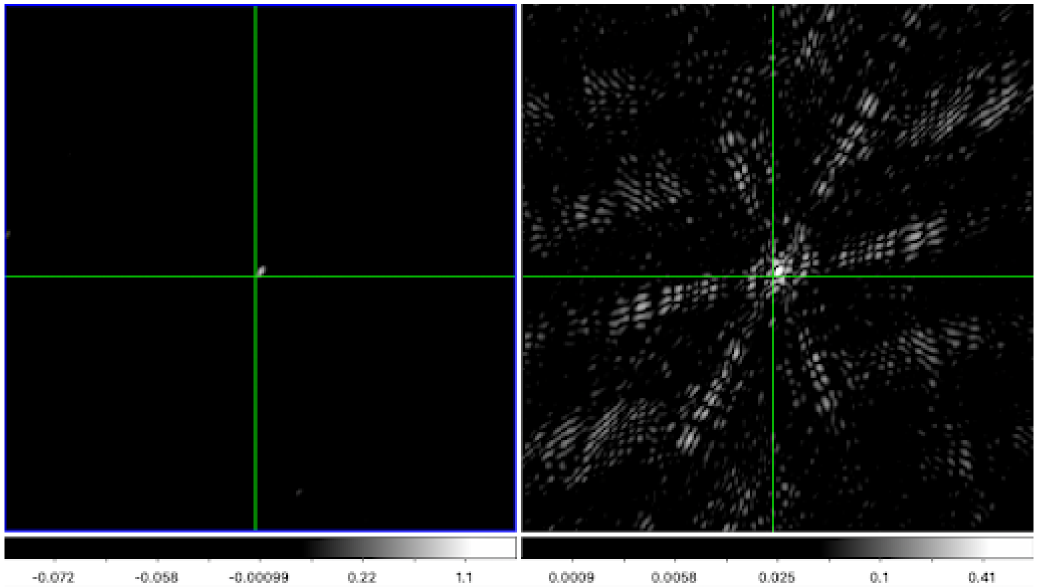

Figure 3: Intensity (left) and polarized (right) images of confirmed LPT ASKAP J183950.5-075635.0 from VLITE's C configuration.

For each source, we also noted any nearby cataloged radio sources in SIMBAD (the Set of Identifications, Measurements, and Bibliography for Astronomical Data) or NED (the NASA/IPAC Extragalactic Database) to determine the credibility of the detection. In addition, we visually inspected the region surrounding the source using the Aladin Sky Atlas, allowing us to rule out sources that were in close proximity to extended structures like supernova remnants or areas dense with young stellar objects (YSOs). These sources are either likely artifacts of the extended source or are otherwise difficult to investigate for transience due to the high levels of nearby noise and variable sources.

3 Results

Through our analysis, no LPT detections have been confirmed. All candidate LPTs were found to be blended sources, sidelobes, or with space objects close enough to be found within the resolution of the VLITE configuration in question. However, throughout the filtering process, we referenced our list against various catalogs of known radio objects and other objects of interest. No matches were found for stars known to have unusually strong magnetic fields, indicating either that any LPTs in this dataset are uncataloged objects or (as may be determined through further research) the transience of LPTs is not caused by abnormally high magnetic fields.

3.1 Sydney Radio Star Catalog

When crossmatching our list of unpolarized sources with the Sydney Radio Star Catalog, we identified two known radio stars potentially seen in VLITE data exhibiting variation in brightness. These stars were HD 164224 and HD 314741. HD 164224 is a chemically peculiar A-type star with strong silicon and chromium spectra, while HD 314741 is a K-type main sequence star [3]. Their S/N values are, respectively, 6.246 and 5.334; though these values are not significant enough to prove a true flare detection, they are not low enough that a true detection can be ruled out entirely, and a preliminary image investigation failed to assist in distinguishing its veracity. Further investigation can be conducted to investigate the variation in brightness for these two stars, and understand the mechanisms behind such a phenomenon.

4 Discussion

Although our filtering methods yielded no new discoveries of long period transients, valuable insights were still gained regarding the narrowing of candidates to find LPTs.

4.1 Fluences

We calculated fluence constraints on our data to gain an understanding of the observation times and energy of candidate LPT bursts, as well as the limitations of our filtering process. Our estimate for the minimum detectable fluence was calculated by the formula:

$$fluence_{min} = noise \times \tau \times S/N_{min} \times PB \tag{1}$$

Here, τ represents the total amount of time on sky across all the candidate images, S/N_{min} is the signal to noise ratio cut used to filter the data, and PB represents a factor due to primary beam attenuation from VLITE [A], which we estimated as two for all configurations.

We observed median detectable fluences of 0.5 Jy·min for both B and C configurations.

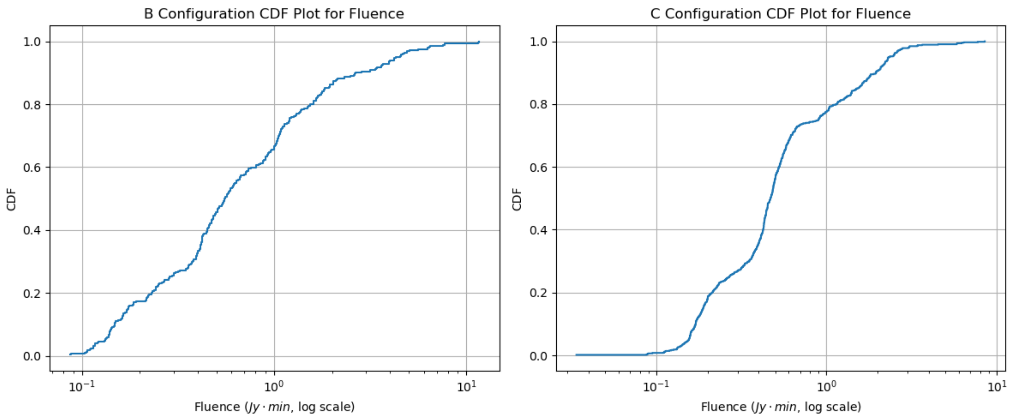

Figure 4: Cumulative distribution functions (CDFs) depicting the fraction of images below a given fluence.

There were two known LPTs in our datasets, ASKAP J183950.5-075635.0 in the C configuration data and GPM J1839-10 in the B configuration data [12, 14]. Not only did we identify them within the data, our filtering process identified every separate image in which they appeared, demonstrating the temporal precision of our process. Our discovery of all known LPTs in the field of view serves as a preliminary validation of our data analysis method and use of visual inspection.

We conducted a Poisson regression analysis to analyze the areal burst density of our LPTs. By understanding the burst density, we can make quantify the chance of finding LPTs in the view of a radio array such as VLITE aboard the VLA. We found that for bright bursts, density is likely $\tilde{1}$ 10^{-5}, indicating that in about 100,000 deg^2 of a radio survey we should expect to find a single bright burst.

Additionally, the scope of our study was sensitive enough that we would have detected an LPT burst of fluence 0.5 Jy·min or greater anywhere in our data's field of view for the B and C Configurations. Future work involving the discovery of long-period transients can use this fluence measure, as well as the total amount of time in the sky observed with unique image IDs. These statistics can help create searches and sky surveys which are optimized to discover LPTs using the methods outlined in this work.

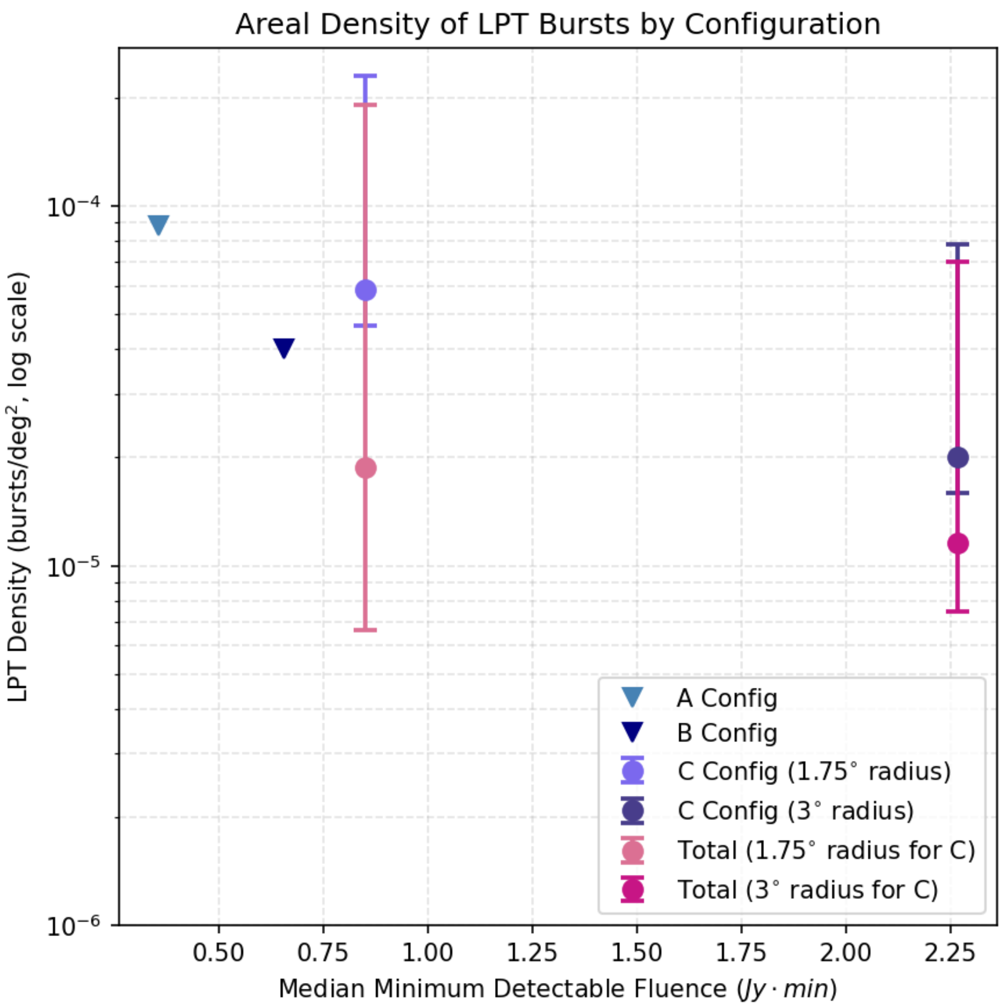

Figure 5: Expected LPT burst densities for various configurations of VLITE.

5 Acknowledgements

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Appendices

A VLITE and Interferometry

VLITE is a system complementary to the VLA, installed on 18 of 27 VLA antennae, that mostly measures in the P-band, or 320–384 MHz [5]. Its receivers are able to collect this low-frequency data at the same time as the VLA is conducting its higher-frequency observations (typically above 500 MHz), allowing for synchronous observation in different radio bands. The VLA itself, located in New Mexico, switches between four different inverted Y-shaped configurations (A, B, C, and D) every few months, each of which has different baseline lengths and thus different resolutions [10]. VLITE takes advantage of a second subreflector to create a focus separate from the primary focus on a VLA dish, which is what allows for the simultaneous data collection that makes it so useful and provides much of the archival data we plan to examine.

VLITE processes its data using a "virtual focus", which allows multiple source readings to be combined as if they were taken over one large parabolic dish in a process called aperture synthesis, seen in Figure 7. [16]

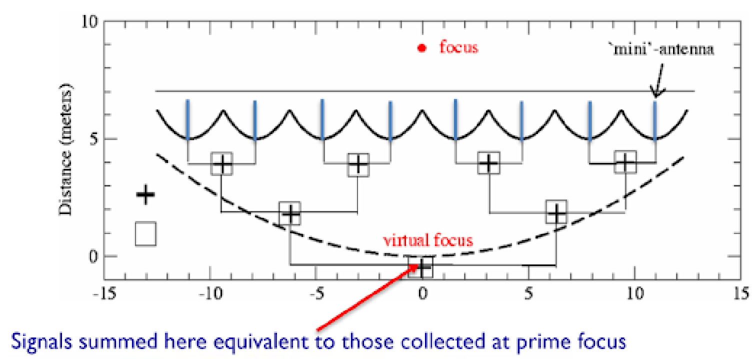

Figure 6: Virtual focus system is shown, demonstrating how multiple antennae work together to mimic the radio data collection of a single large dish.

The longer the baseline, the more accurate the interference pattern the measurement yields will be. This property is the reason behind the importance of aperture synthesis; while it is impractical to build a 100km dish for high-resolution data, it is far simpler to create a virtual dish 100km in size to accomplish the same task.

The electric field data measured by VLITE is converted to a complex-valued voltage that preserves the phase of the incoming light - the average of the product of the voltages yields some value that corresponds to how close the light waves were to being in phase [16]. These averages are taken over short intervals of time and are used to generate interference patterns for the incoming light [10].

The total power received P is proportional to the sky brightness $I(l, m)$ convolved with (multiplied by) a function $A(l, m)$, integrated over the angular area of the beam:

$$P \propto \int I(l, m) A(l, m)\, d\Omega \tag{2}$$

where l and m are coordinates representing the position of the source, and defined with respect to the angle α between the direction the antenna is pointing and the direction of the signal being considered in analysis. l is defined as $sin(\alpha)$ while m is defined as $cos(\alpha)$, giving a helpful frame of reference for the source's location relative to the antenna orientation. This allows the brightness to be isolated from the antenna's beam pattern more easily, as the influence of the beam pattern on the signal depends on the signal's position relative to the antenna's center.

To recover brightness information at a given set of angular coordinates, as is useful for image creation, a function called the visibility function is used. The visibility is the Fourier transform of the brightness; it is analyzed on the plane created by u and v, which represent the x- and y-components (respectively) of the baseline between antennae in terms of the observation wavelength. The function is created from two complex voltages

$$R_C = P\cos(\omega\tau_g)$$
$$R_S = P\sin(\omega\tau_g)$$

where P is the received power ($\frac{E^2}{2}$, where E is the voltage), and τ_g is the geometric delay. These two voltages together form the basis of the Fourier transform (which takes the form of $R_C - iR_S$)[16].

The visibility itself is represented by the following equation:

$$V(u, v) = \iint I(l, m) e^{-i2\pi(ul+vm)}\, dl\, dm \tag{3}$$

Using the coordinates attached to a certain visibility measurement, the brightness data is recovered through the reverse Fourier transform, and is ultimately used to generate images.

B Data Processing

Using data from known radio surveys, we plotted nearest neighbors to determine distance cutoffs, ensuring that we were investigating only undiscovered sources in our data.

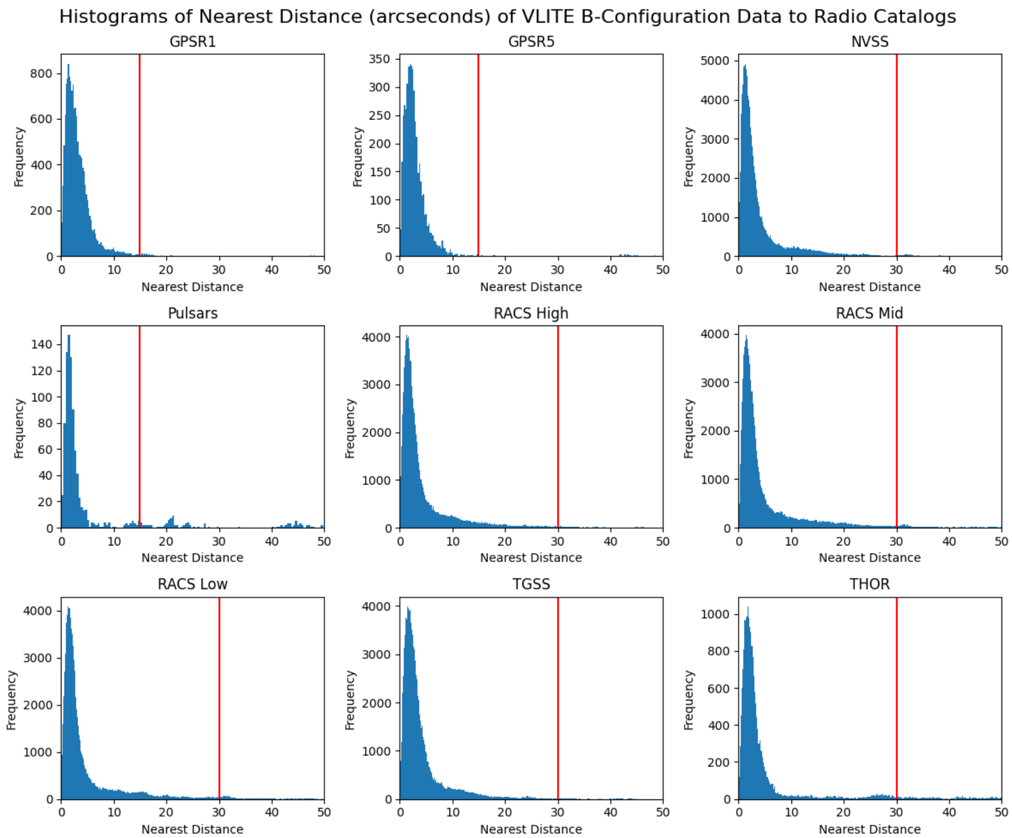

Figure 7: Nearest neighbor visualizations from GSPR1, GSPR5, NVSS, RACS Low, Middle, and High, TGSS, THOR, and known pulsars.

From the plots, we applied the following cutting distances, in arcseconds: 45 for NVSS; 30 for TGSS, THOR, RACSlow, RACSmid, RACShigh; 15 for known pulsars; and 15 for GSPR1 and GSPR5.

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